Water Ledger Michigan: methods brief

Model version 1.2 · data snapshot 2026-10-29 · waterledger.org · github.com/northlinehydrology/waterledger

What it estimates

For any city or township in Michigan, Water Ledger estimates how much water a data center campus of a given size and cooling design would take, use up and send back, on an average day and on a 1-in-10-year peak day, and sets that beside the public water system, the sewer plant, the streams downstream and the electric utility that would serve it. It uses public data only and is a screening estimate, not a design.

Data

Category	Source	In Michigan
Places	Census TIGERweb: county subdivisions (Michigan's cities and townships)	1,541 places
Water supply	EPA community water system service areas, with SDWIS population and connections	411 systems
Wastewater	EPA Clean Watersheds Needs Survey 2022 (design flow) matched to ECHO (recent flow, receiving water)	394 plants
Receiving waters	USGS Watershed Boundary Dataset; EPA ATTAINS (21MICH, 2026 report)	1,085 watersheds with a downstream path; 1,354 watersheds on those paths
Climate	ERA5 reanalysis (Open-Meteo), 1995-2024, at eight Michigan regions; Atlas 14 design storms; Climate at a Glance rainfall	8 regions
Energy	MPSC territories; EIA-861 sales and peaks	57 utilities with loads; grid regions: MISO, PJM

How an estimate is made

Cooling water. Hour by hour over 30 years of weather, the campus's heat load is turned into evaporation, drift and blowdown for each cooling design: cooling towers, hybrid (evaporative only in warm weather), closed loop with evaporative assist on the hottest hours, and closed loop with no water for cooling. The evaporative share follows Rutberg et al. (2011); blowdown follows the DOE FEMP water balance.

Uncertainty. The model runs 200 times with its 15 cooling parameters drawn from published or estimated ranges (marked [A] where they stand in for a facility's own data). Results are the 10th, 50th and 90th percentiles; the peak day is the 90th percentile of each year's highest day.

Staff water and loop top-ups are added at the campus's size (people on site, and small leaks from closed loops).

Source and wastewater. The campus uses the public water system mapped in the place; with none mapped, a closed loop uses the nearest system serving over 3,300 people if it is within 10 km, and anything else uses new wells. Wastewater goes to the sewer when a plant is in reach and, for evaporative designs, has room for the blowdown; otherwise a septic system (closed loop) or the campus's own discharge.

Downstream. Each place's main watershed is followed downstream (up to 25 watersheds, across state lines), with the impairments and cleanup plans (TMDLs) EPA lists for each.

What Michigan adds

Michigan is the full version: its own climate regions, EGLE's reported water use by county and sector, household wells (EGLE Wellogic), designated trout streams (Michigan DNR), lake-source cooling on the Great Lakes, and Michigan's rules (Part 327 withdrawals, Part 22 septic limits, Part 91 soil erosion, Rule 57 and temperature standards). A data center proposal and moratorium tracker is kept by hand on waterledger.org.

Example: 300 MW of IT capacity in Gaines Charter Township

Million gallons a day: median, with the 10th to 90th percentile range in brackets. Peak day is the 1-in-10-year highest day. "Used up" is water evaporated or otherwise not returned. Defaults for water source and wastewater are filled in by the rules above (muni: public system; sewer: the nearest plant).

Design	Withdrawn, average day	Withdrawn, peak day	Used up, average day	Water from / to
Cooling towers	2.34 (1.91-2.70)	2.94 (2.40-3.41)	1.87 (1.56-2.16)	muni / sewer
Hybrid	0.10 (0.05-0.16)	1.94 (1.55-2.45)	0.07 (0.04-0.13)	muni / sewer
Closed loop with assist	0.01 (<0.01-0.01)	0.25 (0.15-0.40)	<0.01 (<0.01-0.01)	muni / sewer
Closed loop	<0.01	<0.01	<0.01	muni / sewer

Checks on this snapshot

Michigan's snapshot carries the site's own files; tests confirm its answers are identical either way, and that the shared files agree with Michigan's for places chosen at random.

Limits

These are screening estimates from public data, for comparing designs and sizes before an applicant's numbers exist. They do not replace a facility's design data, a utility's capacity analysis or a permit review. The watershed path is followed watershed to watershed, not stream by stream. EPA assessments reflect each state's latest report. Weather is the past 30 years.

Planning-level hydrologic assessment. Not engineering design. Built with the open waterledger Python packages (waterledger 2.0.0); methods, parameters and sources: waterledger.org and the package repository. Northline Hydrology.